3

SECTION 2: PROPOSAL OF POTENTIAL LANDSLIDE MITIGATION MEASURES

3.1 Introduction

Because of the variety of the landsides processes, the nature of regolith and rock environments and landscapes in which they are found (Appendix 1), virtually every slope mitigation design problem is unique. Designing a stable slope includes field investigations, laboratory tests, stability analyses, and proper construction control. Because most of the details involved in such a work cannot be standardized, good engineering judgment, experience, and intuition must be coupled with the best possible data gathering and analytical techniques to achieve a safe and economical solution to slope stabilization (Turner and Schuster, 1996).

The goal of this section is to provide a general overview of the potential mitigation measures that could be tested for the areas identified in steps 1 to 5 as at risk and vulnerable to landslides. With the first paragraph in mind that any mitigation measures should be implemented based on data and information that we have not at this stage of our knowledge, one have to keep in mind that the purpose of this section is not to go beyond the sole role of the literature review. Further research and expertise actions are needed if one want to move towards the implementation of concrete mitigation measures. The measures presented here are therefore not catchment-specific, and to some extent not specific to a type of landslide in particular. The structure of the report is as follow: first, the mitigation measures are detailed. Along with criteria for the selection of the mitigation measures, details are provided on measures at the level of the hazard, the vulnerability, and the element at risk. The measures are then discussed in the context of the Catchment Restoration Opportunity Mapping Support System (CROM DSS). Together with the presentation of the mitigation measures, additional figures and appendixes explaining and illustrating key concepts are proposed to ease the understanding of the document. As a last note, a brief focus is provided on gully erosion. Although not landslide processes, problems of gullying have been identified in the study areas.

Figure 10 summarizes the framework for landslide risk management. This represents a framework widely used internationally (Fell et al., 2008). In this document on potential mitigation measures, we focus on one of the aspects of the whole risk management process.

Before starting presenting the potential mitigation measures that could be tested in the four study areas, it is important to provide an overview of the main findings of the first part of the analysis that concerned the landside risk and that are relevant for the mitigation focus. In this first part of the analysis were identified several slope failure processes: